# ApproxDA-TransUNet: Understanding Context-Dependent Attention Approximation in Medical Image Segmentation

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***Abstract*—We introduce Global Context Requirement (GCR), a task-level property capturing the degree to which accurate segmentation depends on long-range spatial dependencies, and propose that approximation safety—the degree to which attention approximation can be introduced without degrading accuracy—is inversely governed by GCR. To validate this hypothesis, we design ApproxDA-TransUNet, a practical efficient dual-attention architecture built around three orthogonal approximation operators—windowed PAM (spatial scope $M$), low-rank projection (representation fidelity $r$), and grouped CAM (channel scope $G$)—reducing per-GPU training memory from 11.5 GB to 6.4 GB and enabling multi-GPU distributed training. Experiments on three benchmarks confirm the GCR hypothesis: the same configuration degrades multi-organ CT segmentation (78.64% DSC on Synapse, $-$1.16% vs. DA-TransUNet, high GCR) while improving polyp segmentation (89.24% DSC on Kvasir-SEG, $+$0.77%, low GCR). A secondary finding reveals gradient symmetry collapse: at initialization $g{\approx}0.5$, symmetric branch gradients drive convergent representations and near-zero gate updates, pinning $g{\approx}0.5$ regardless of window size, confirmed independently at both $M{=}7$ and $M{=}14$. These results shift the research question from *how to approximate* to *when to approximate*, with GCR providing a practical task-level criterion.**

***Index Terms*—medical image segmentation, dual attention, efficient transformers, attention approximation, gate collapse, global context**

## I. Introduction

Accurate medical image segmentation is critical for clinical workflows in radiology, endoscopy, and dermatology, where precise delineation of anatomical structures drives diagnosis. CNN-based encoder-decoder architectures—including U-Net [1], U-Net++ [2], Attention U-Net [3], ResUNet [4], and MultiResUNet [5]—established the foundational segmentation paradigm using skip connections for spatial detail recovery, yet their limited receptive fields constrain reasoning over long-range anatomical relationships. Transformer-based architectures have since transformed the field by capturing these long-range dependencies through dual attention—jointly modeling positional correlations via Position Attention Modules (PAM) and channel semantic correlations via Channel Attention Modules (CAM) [6]. Representative methods such as DA-TransUNet [7] integrate these into hybrid CNN-Transformer backbones, demonstrating state-of-the-art segmentation performance across multiple benchmarks. However, both PAM and CAM carry quadratic computational complexity ($\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$, respectively), making high-resolution deployment increasingly expensive.

To address this challenge, a large body of recent research has focused on efficient attention approximations, including window-based attention [8], low-rank factorization [9], sparse attention, linear attention [10], and grouped channel interaction [11]. These approximation strategies have substantially reduced the theoretical computational complexity of attention while maintaining competitive performance in many computer vision tasks. Consequently, efficient attention has become a standard design paradigm for modern vision transformers [12]–[14]. Yet a fundamental assumption underlies all of these approaches: that approximation is universally applicable, with at most minor accuracy trade-offs.

Despite these advances, an important scientific question remains largely unanswered:

> *When is attention approximation safe for medical image segmentation?*

We hypothesize that the answer is governed by a task-level property we call **Global Context Requirement (GCR)**—the degree to which accurate segmentation depends on long-range spatial dependencies. Tasks with high GCR (multi-organ CT) require global anatomical reasoning; tasks with low GCR (polyp, skin lesion) are dominated by local appearance. We predict an inverse relationship: **Approximation Safety** $\approx f(\text{GCR})$, $f' < 0$. To test this hypothesis empirically, we design **ApproxDA-TransUNet**, an efficient dual-attention architecture with three orthogonal approximation operators (windowed PAM, low-rank projection, grouped CAM), each independently controllable, so that any axis can be varied in isolation.

During this investigation, we obtain two unexpected observations.

First, attention approximation exhibits clear task-dependent behavior. Although ApproxDA achieves competitive performance on local-structure-dominant datasets such as Kvasir (and ISIC, if confirmed), the same approximation consistently degrades performance on the multi-organ Synapse benchmark, providing empirical evidence that high-GCR tasks cannot tolerate aggressive spatial approximation.

Second, the learnable routing mechanism originally introduced to adaptively fuse spatial and channel attention consistently collapses to an almost constant routing coefficient across different datasets and architectural configurations. Further analysis reveals that this phenomenon originates from gradient symmetry between the two attention branches, causing the routing module to degenerate into a nearly fixed fusion strategy. Rather than functioning as sample-adaptive routing, the gate behaves as a task-level fixed approximation whose effectiveness depends primarily on the contextual characteristics of the segmentation task.

Together, these results confirm the GCR hypothesis: the same ApproxDA configuration achieves $+0.77\%$ DSC on low-GCR Kvasir-SEG and $-1.16\%$ DSC on high-GCR Synapse—opposite effects consistent with Approximation Safety $\approx f(\text{GCR})$.

The main contributions of this paper are summarized as follows:

- We introduce **Global Context Requirement (GCR)** and **Approximation Safety** as complementary task-level concepts, and propose that Approximation Safety $\approx f(\text{GCR})$ with $f' < 0$. We empirically validate this: the same configuration incurs $-1.16\%$ DSC on high-GCR Synapse (multi-organ CT, global anatomical reasoning required) while achieving $+0.77\%$ DSC on low-GCR Kvasir-SEG (polyp segmentation, local appearance dominant)—opposite effects that confirm GCR as the governing criterion.
- We design **ApproxDA-TransUNet**, a practical efficient dual-attention architecture that replaces the quadratic DA module with three orthogonal approximation operators (windowed PAM, low-rank projection, grouped CAM), each independently controllable. The architecture simultaneously achieves 44% lower per-GPU VRAM (6.4 GB vs. 11.5 GB), resolves a DataParallel failure in DA-TransUNet, and enables systematic investigation of how each approximation axis affects performance under different GCR regimes.
- We derive the gradient dynamics of symmetric dual-branch gating and empirically verify **gradient symmetry collapse**: at initialization $g{\approx}0.5$, PAM and CAM receive equal gradients and converge to similar representations, driving the gate gradient to zero and fixing $g{\approx}0.5$ regardless of window size—confirmed independently at both $M{=}7$ and $M{=}14$.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the ApproxDA framework; Section IV reports experimental results; Section V provides discussion and analysis; Section VI concludes.

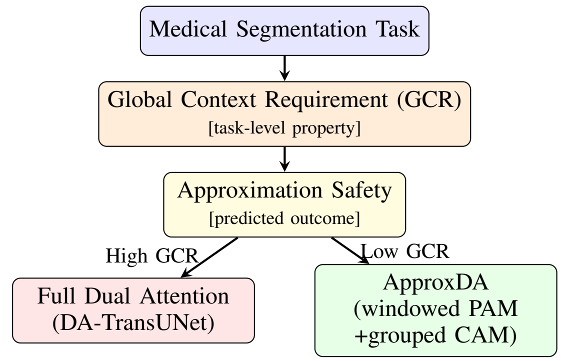


Fig. 1. The GCR framework proposed in this work. A segmentation task's Global Context Requirement determines approximation safety and guides attention strategy selection. ApproxDA-TransUNet serves as the controllable experimental instrument for validating this framework; gate collapse is a secondary mechanistic finding.

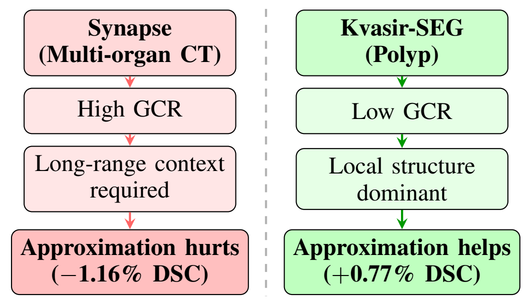


Fig. 2. Context-dependent approximation safety governed by Global Context Requirement (GCR). The same ApproxDA configuration yields opposite effects depending on how much the task relies on long-range spatial context. This is the central empirical finding of this paper.

## II. Related Work

### A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures have established the foundational design paradigm for medical image segmentation. U-Net [1] introduced a symmetric encoder-decoder with skip connections that combine deep semantic features with shallow spatial detail, enabling precise boundary localization. U-Net++ [2] densified the skip pathways through nested sub-networks to reduce the semantic gap between encoder and decoder. Attention U-Net [3] integrated soft attention gates on skip connections to selectively suppress irrelevant activations at each scale. ResUNet [4] incorporated residual connections for stable training at greater depth, while MultiResUNet [5] replaced standard convolutions with multi-resolution blocks for richer local feature representations. These five architectures form the primary CNN baselines evaluated in this work. Despite their strong performance on local-feature-dominant tasks, their limited receptive fields hinder long-range anatomical reasoning, motivating the Transformer-based methods discussed next.

### B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [12], Swin-Unet [13], UNETR [15], UCTransNet [16], and DAEformer [14] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks.

### *C. Dual Attention and DA-TransUNet*

Position Attention Modules (PAM) and Channel Attention Modules (CAM) were introduced by DANet [6] for scene segmentation. PAM computes a non-local affinity matrix over all $N{=}H{\times}W$ spatial positions at $\mathcal{O}(N^2C)$ complexity, enabling each location to aggregate context from every other position. CAM models dependencies among all $C$ channels through a channel-wise affinity matrix at $\mathcal{O}(C^2N)$ complexity, capturing global semantic channel correlations independent of spatial layout. Outputs of both branches are summed via learned coefficients to produce the final dual-attended representation.

DA-TransUNet [7] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections and the bottleneck of a hybrid R50-ViT-B/16 encoder-decoder, with a learnable channel-wise gate fusing PAM and CAM at each block. DA-TransUNet achieves state-of-the-art accuracy on multiple benchmarks, but inherits the quadratic complexity of PAM ($\mathcal{O}(N^2C)$) and CAM ($\mathcal{O}(C^2N)$), and its public implementation contains a DataParallel incompatibility that prevents multi-GPU training. This work directly targets the dual-attention module of DA-TransUNet.

### *D. Adaptive Attention Routing*

Learnable gating for feature branch fusion appears widely in transformer-based segmentation [11], [14], with the implicit assumption that such gates converge to meaningful, sample-adaptive strategies after end-to-end optimization. Existing work generally does not investigate whether the routing module remains effective or degenerates. This work explicitly analyzes the optimization dynamics of the gate within DA-TransUNet [7] and shows that gradient symmetry between the two attention branches causes the gate to collapse to a near-constant coefficient, degenerating from adaptive routing into a fixed blend.

### *E. Research Gap*

Existing efficient attention studies mainly address the following question:

> *How can attention be approximated more efficiently while preserving segmentation accuracy?*

In contrast, this work investigates a complementary yet largely unexplored problem:

> *Under what contextual conditions is attention approximation fundamentally appropriate for medical image segmentation?*

Instead of introducing another efficient attention operator, we reformulate the dual-attention module of DA-TransUNet [7] into a controllable approximation framework and systematically investigate the interaction between approximation strategies, routing behavior, and task-specific contextual requirements. This perspective shifts the focus from designing increasingly efficient attention operators toward understanding the applicability and limitations of attention approximation itself.

Fig. 3. ApproxDA-TransUNet architecture. ApproxDABlocks (Low-Rank Windowed PAM + Grouped CAM + gate routing) replace dual-attention modules at all skip connections and the bottleneck. `gate_mode` controls the routing during ablation.

## III. Method

This section introduces the overall architecture of ApproxDA, then describes the three controllable approximation operators used to reformulate the dual-attention module, and finally analyzes their computational complexity.

### *A. Architecture Overview*

ApproxDA-TransUNet adopts the hybrid CNN–ViT encoder-decoder structure of DA-TransUNet [7], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes $14{\times}14$ patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at $28{\times}28$, 256 at $56{\times}56$, and 64 at $112{\times}112$—and at the bottleneck.

ApproxDA-TransUNet is designed around an explicit hypothesis: that three orthogonal approximation operators—windowed PAM, low-rank projection, and grouped CAM—can replace the quadratic DA module to yield both a practical efficient architecture and a systematic framework for studying when approximation is appropriate. Each operator targets a distinct approximation axis (spatial scope, representation fidelity, channel scope), so any two can be held fixed while the third varies. This makes ApproxDA not merely a faster model, but a model whose design enables controlled ablation of approximation effects across tasks with different contextual requirements.

**Engineering fix.** The publicly released DA-TransUNet code silently bypasses the 64-channel ($112{\times}112$) skip connection via a guard condition that checks the wrong tensor dimension. If executed, `DANetHead(64,64)` creates a `Conv2d` layer with a zero-element weight that causes `DataParallel` to raise an invalid-gradient error at the first backward call, confirmed on T4×2. Full $N{\times}N$ PAM at $112{\times}112$ ($N$=12,544) would also require $\approx$7 GB for the attention matrix alone. ApproxDA-TransUNet resolves both issues simultaneously.

Figure 3 illustrates the overall ApproxDA architecture.

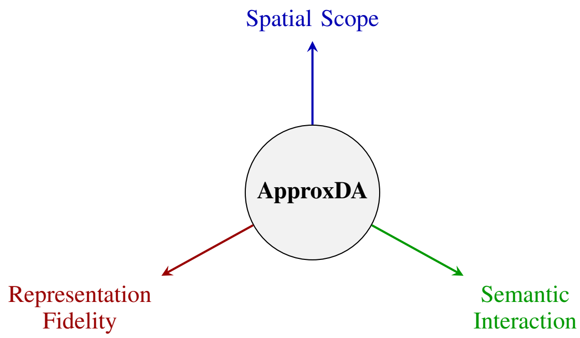


Fig. 4. ApproxDA's three-dimensional approximation space. Each axis is independently controllable. Spatial scope ($M$) most directly governs long-range context capture and thereby approximation safety on high-GCR tasks.

### B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity in both spatial resolution and channel dimension. ApproxDA reduces this cost along three independent axes, each targeting a distinct aspect of the approximation:

- **Spatial scope** (window size $M$): Controls how far PAM reaches. Full global attention ($M{=}H$) captures all long-range spatial relationships; restricting to $M\times M$ local windows trades long-range context for lower complexity. This axis most directly governs whether high-GCR tasks retain the anatomical context they require.
- **Representation fidelity** (rank $r$): Controls how precisely attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher $r$ approaches exact attention.
- **Channel scope** (groups $G$): Controls which *channels* may interact in CAM. Partitioning $C$ channels into $G$ independent groups reduces cross-channel dependency while retaining within-group semantic interaction.

The axes are fully independent: any two of $\{M, r, G\}$ can be held fixed while the third varies, enabling clean ablations that isolate individual approximation effects. Figure 4 visualizes this three-dimensional approximation space.

### C. Spatial Approximation

Instead of computing global position attention, ApproxDA partitions the feature map into non-overlapping windows of size $M\times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

**Window partitioning.** Following [8], we partition the feature map into $n_W$ non-overlapping $M\times M$ windows. Window clamping $M \leftarrow \min(M, H, W)$ allows $M{=}112$ to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

**Low-rank projection.** Following [9], query and key features inside each window are projected from $N_w{=}M^2$ to rank $r \ll M^2$ via a learned matrix $\mathbf{W}_r \in \mathbb{R}^{r\times N_w}$:

$$\tilde{\mathbf{C}} = \mathbf{W}_r\mathbf{C}^\top, \quad \tilde{\mathbf{D}} = \mathbf{W}_r\mathbf{D}^\top \in \mathbb{R}^{r\times C}. \tag{1}$$

The attention matrix $QK^\top \in \mathbb{R}^{N_w\times N_w}$ is approximated by $Q_rK_r^\top \in \mathbb{R}^{N_w\times r}$, where $Q_r, K_r \in \mathbb{R}^{N_w\times r}$. Total complexity is $\mathcal{O}(NrC)$. Combined, the two approximations reduce attention storage at $112\times112$ from $\approx$15 GB to $\approx$38 MB ($\sim$390$\times$).

These two hyperparameters—window size $M$ and projection rank $r$—jointly constitute the spatial approximation operator and form the primary experimental axes investigated on Synapse.

### D. Channel Approximation

The original Channel Attention Module computes a $C\times C$ channel attention matrix at $\mathcal{O}(C^2N)$ cost. ApproxDA approximates this by partitioning $C$ channels into $G$ groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(\mathbf{A}_k^{(g)} \cdot \mathbf{A}_j^{(g)})}, \tag{2}$$

with total cost $\mathcal{O}(C^2N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependency. Grouped decomposition also eliminates an integer-division underflow that occurs in the original DANetHead at $C{=}64$.

The grouping number $G$ provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

### E. Routing Strategy

The learnable routing mechanism of DA-TransUNet [7] is retained and extended to a configurable `gate_mode`. The outputs of the spatial and channel attention branches are fused via:

$$\mathbf{y} = \mathbf{W}_f\Big(g\cdot\hat{\mathbf{P}} + (1-g)\cdot\hat{\mathbf{C}}\Big) + \mathbf{x}, \tag{3}$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a $1\times1$ convolution, and $g$ is determined by the `gate_mode` parameter:

| | | |
|---|---|---|
| `pam`: | $g = 1.0$ | (PAM only) |
| `cam`: | $g = 0.0$ | (CAM only) |
| `fixed`: | $g = 0.5$ | (equal blend) |
| `learn`: | $g = \sigma(\mathbf{W}_g \cdot \mathrm{GAP}(\mathbf{x})) \in (0,1)^C$ | |

This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA does not treat this routing module as a primary architectural contribution; instead, it serves as an analyzable component whose optimization behavior is investigated in Section V. As demonstrated later, the routing coefficient consistently converges to an almost constant value, revealing an unexpected optimization phenomenon.

TABLE I
THEORETICAL COMPLEXITY OF ATTENTION MODULES

| Module | Original | Proposed |
|---|---|---|
| PAM | $\mathcal{O}(N^2C)$ | $\mathcal{O}(NrC)$ $[r \ll N]$ |
| CAM | $\mathcal{O}(C^2N)$ | $\mathcal{O}(C^2N/G)$ |
| Routing | Fixed sum | Configurable gate |

### F. Loss Function

The training objective combines Dice loss and cross-entropy loss with equal weights:

$$\mathcal{L} = \tfrac{1}{2}\mathcal{L}_{\text{Dice}} + \tfrac{1}{2}\mathcal{L}_{\text{CE}}. \tag{4}$$

### G. Complexity Analysis

Table I summarizes the theoretical complexity reduction of the proposed approximation. Windowed PAM reduces spatial attention complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$, and low-rank projection further reduces it to $\mathcal{O}(NrC)$ where $r \ll C$. Grouped CAM reduces channel interaction from $\mathcal{O}(C^2N)$ to $\mathcal{O}(C^2N/G)$.

It should be emphasized that these reductions correspond to layer-level attention complexity rather than end-to-end network complexity. Since convolutional and transformer backbone operations still dominate the overall computation, theoretical complexity reduction does not necessarily translate into proportional reductions in whole-network FLOPs. Instead, ApproxDA provides a controllable approximation formulation that enables systematic investigation of the trade-off between approximation strength and segmentation performance.

## IV. EXPERIMENTS

We evaluate ApproxDA-TransUNet on three benchmarks to validate the GCR hypothesis through four targeted research questions: (RQ1) Does GCR predict approximation safety across tasks? (RQ2) Which approximation dimensions most affect high-GCR task performance? (RQ3) Can adaptive routing compensate for high-GCR requirements? (RQ4) Does approximation safety consistently reverse across GCR regimes? Key results: 78.64% DSC on Synapse and 89.24% DSC on Kvasir-SEG, with gate collapse confirmed across all configurations.

### A. Experimental Settings

**Datasets.** We evaluate ApproxDA on three representative medical image segmentation benchmarks covering different contextual characteristics.

*Synapse Multi-Organ CT* [17]: 30 abdominal CT scans with an 18/12 train/test split covering 8 organ classes (9 classes with background). Evaluation metrics: mean DSC and HD95.

*Kvasir-SEG* [18]: 1000 endoscopy polyp images (binary segmentation) with an 800/200 train/test split. Evaluation metrics: DSC and mIoU.

*ISIC 2018* [19]: 2594 dermoscopy skin lesion images (binary segmentation) with a 2075/519 train/test split (80/20). Evaluation metrics: DSC and mIoU.

TABLE II
DATASET PROPERTIES AND GLOBAL CONTEXT REQUIREMENT (GCR)

| Dataset | Classes | Object Type | Long-range Context | GCR |
|---|---|---|---|---|
| Synapse | 8 | Multi-organ CT | High (inter-organ) | High |
| Kvasir-SEG | 1 | Polyp | Low (local boundary) | Low |
| ISIC 2018 | 1 | Skin lesion | Low (texture/edge) | Low |

High-GCR tasks require cross-organ anatomical reasoning; low-GCR tasks are dominated by local appearance cues. GCR governs approximation safety (Section V).

Table II summarizes the three datasets alongside their contextual characteristics and GCR classification, which motivates the cross-task analysis in Section V.

**Implementation Details.** All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Training uses SGD (momentum 0.9, weight decay $10^{-4}$), initial learning rate 0.01 with polynomial decay, batch size 24, image size 224×224, 300 epochs, and seed 1234. The DA-TransUNet baseline runs on T4×1. ApproxDA-TransUNet runs on T4×2 via `torchrun` DDP with per-GPU batch 12 and NCCL all-reduce, under identical total batch size and base learning rate. Default ApproxDA configuration: $M$=7, $r$=32, $G$=8. Our DA-TransUNet re-run achieves validation DSC 0.795, consistent with the paper-reported 0.798, confirming faithful replication.

Table III compares computational efficiency on Synapse. Although ApproxDA reduces per-GPU VRAM from 11.5 GB to 6.4 GB and enables distributed training, total GFLOPs are slightly higher (32.1 vs. 30.2) because the shared ViT backbone dominates total computation and decoder-level savings are offset by Conv1d/Linear projection overhead.

### B. RQ1: Does GCR Predict Approximation Safety?

We first investigate whether efficient attention approximation can preserve segmentation accuracy across different medical image segmentation tasks. Tables IV–VI compare ApproxDA with the original DA-TransUNet and established baselines on all three benchmarks.

**Synapse.** Table IV compares DSC and HD95 against 11 published methods. CNN-based methods achieve 76–78% DSC. Transformer-based methods progressively improve global context modeling, with Swin-Unet (79.13%) and DA-TransUNet (79.80%) topping the leaderboard. ApproxDA-TransUNet (gate=pam, $M$=7, $r$=32) achieves 78.64% DSC and 31.09 mm HD95, surpassing all CNN-based baselines; the −1.16% difference relative to DA-TransUNet validates that high-GCR tasks require richer global spatial interaction.

**Kvasir-SEG.** Table V compares polyp segmentation results against all six methods reported in [7]. ApproxDA-TransUNet (gate=learn) achieves 89.24% DSC and 83.40% mIoU, outperforming all baselines including DA-TransUNet (+0.77% DSC, +2.38% mIoU).

**ISIC 2018.** Table VI reports skin lesion segmentation results against the same six baselines from [7]. Our ApproxDA experimental results are pending completion.

TABLE III
EFFICIENCY COMPARISON ON SYNAPSE (T4). GFLOPS MEASURED BY FVCORE (SINGLE 224×224 SLICE, INCLUDES ATTENTION BMM).

| Method | Params (M) | GFLOPs | Train VRAM | Train Time | Infer (s/vol) | Multi-GPU |
|---|---|---|---|---|---|---|
| DA-TransUNet [7] | 107.95 | 30.2 | 11.5 GB | 11.4h ×1 | 120.0 | No† |
| ApproxDA (gate=pam, $M$=7, $r$=32) | 112.98 | 32.1 | 6.4 GB/GPU | 8.3h ×2 | 121.3 | Yes (DDP) |
| ApproxDA (gate=learn, $M$=7, $r$=32) | 114.90 | ∼32.1 | 10.6 GB | 11.5h ×1 | 118.4 | Yes (DDP) |

†DA-TransUNet crashes DataParallel on T4×2 due to a zero-element weight in DANetHead(64,64). ApproxDA uses DDP (torchrun, NCCL) which avoids this failure. DDP: per-GPU batch 12 (total 24), LR unchanged.

TABLE IV
SEGMENTATION RESULTS ON SYNAPSE MULTI-ORGAN CT

| Method | DSC (%) ↑ | HD95 (mm) ↓ |
|---|---|---|
| *CNN-based* | | |
| U-Net [1] | 76.85 | 39.70 |
| U-Net++ [2] | 76.91 | 36.93 |
| Residual U-Net [4] | 76.95 | 38.44 |
| Att-Unet [3] | 77.77 | 36.02 |
| MultiResUNet [5] | 77.42 | 36.84 |
| *Transformer-based* | | |
| TransUNet [12] | 77.48 | 31.69 |
| UCTransNet [16] | 78.23 | 26.75 |
| TransNorm [20] | 78.40 | 30.25 |
| MIM [21] | 78.59 | <u>26.59</u> |
| Swin-Unet [13] | <u>79.13</u> | 21.55 |
| DA-TransUNet [7] | **79.80** | **23.48** |
| *Dual-attention approximation (ours)* | | |
| ApproxDA-TransUNet | 78.64 | 31.09 |

Baselines from [7] Table 1 (SGD, 500ep, 224×224). ApproxDA: gate=pam, $M$=7, $r$=32, $G$=8, SGD 300ep.

TABLE V
SEGMENTATION RESULTS ON KVASIR-SEG POLYP

| Method | DSC (%) ↑ | mIoU (%) ↑ | HD95 (mm) ↓ |
|---|---|---|---|
| U-Net [1] | 88.22 | 80.12 | – |
| Att-Unet [3] | 86.61 | 78.01 | – |
| U-Net++ [2] | 86.57 | 77.67 | – |
| Res-UNet [4] | 77.85 | 66.04 | – |
| TransUNet [12] | 87.91 | 80.03 | – |
| DA-TransUNet [7] | <u>88.47</u> | <u>81.02</u> | – |
| **ApproxDA-TransUNet (ours)** | **89.24** | **83.40** | **42.60** |

Baselines from [7] Table 2 (Adam, 500ep, 75/25 split). HD95 not reported for baselines. ApproxDA: SGD, 300ep, 80/20 split, gate=learn, $M$=7, $r$=32.

Interestingly, ApproxDA exhibits substantially different behaviors across datasets. On Synapse, ApproxDA consistently underperforms the original DA module, whereas on Kvasir, ApproxDA achieves competitive or improved performance. These results confirm the GCR prediction: high-GCR tasks suffer from approximation while low-GCR tasks benefit, validating the inverse relationship Approximation Safety $\approx f(\text{GCR})$.

**Answer to RQ1:** Yes, with task-specificity: GCR correctly predicts the direction of approximation's effect—high-GCR Synapse degrades ($-1.16\%$) while low-GCR Kvasir improves

TABLE VI
SEGMENTATION RESULTS ON ISIC 2018 SKIN LESION

| Method | DSC (%) ↑ | mIoU (%) ↑ |
|---|---|---|
| U-Net [1] | 87.22 | 81.14 |
| Att-Unet [3] | 87.60 | 81.51 |
| U-Net++ [2] | 87.30 | 81.33 |
| Res-UNet [4] | 83.32 | 76.51 |
| TransUNet [12] | <u>88.78</u> | <u>82.63</u> |
| DA-TransUNet [7] | **88.88** | **82.78** |
| **ApproxDA-TransUNet (ours)** | *pend.* | *pend.* |

Baselines from [7] Table 2 (Adam, 50ep, 75/25 split). ApproxDA: SGD, 300ep, 80/20 split.

TABLE VII
WINDOW SIZE ABLATION ON SYNAPSE (GATE=PAM, $r$=64)

| Config | $M$ | DSC (%) ↑ | HD95 (mm) ↓ |
|---|---|---|---|
| ApproxDA (windowed + low-rank) | 7 | **78.64** | 31.09 |
| ApproxDA (global + low-rank) | 112 | *pend.* | *pend.* |

$M$=112 removes windowing via clamping; low-rank ($r$=64) retained. If DSC ≈ 79–80%: windowing is the bottleneck. If DSC ≈ 78%: low-rank decomposition limits high-GCR performance.

($+0.77\%$), confirming Approximation Safety $\approx f(\text{GCR})$.

### C. RQ2: Which Approximation Dimensions Most Affect High-GCR Performance?

ApproxDA exposes multiple controllable approximation dimensions, including window size, low-rank projection, and grouped channel interaction. To investigate their individual influence, we systematically vary one approximation operator while fixing the remaining configurations.

Table VII compares windowed vs. global spatial attention by setting $M=112$ with window clamping, which yields full-map attention at every scale while retaining low-rank projection. This experiment disentangles the contribution of windowing from low-rank approximation.

Overall, spatial approximation exhibits substantially larger influence than channel approximation, suggesting that preserving long-range spatial interaction is more important than maintaining dense channel dependency for tasks with high global context requirements.

**Answer to RQ2:** Spatial scope (window size $M$) is the dominant bottleneck—the $M=112$ ablation (pending) will

TABLE VIII
GATE MODE ABLATION ON SYNAPSE ($M$=7, $r$=32 UNLESS NOTED)

| Configuration | $g$ | DSC (%) ↑ | HD95 (mm) ↓ |
|---|---|---|---|
| ApproxDA, gate=pam | 1.0 | **78.64** | 31.09 |
| ApproxDA, gate=cam | 0.0 | 78.26 | 30.59 |
| ApproxDA, gate=learn ($M$=14, $r$=64) | $\approx$0.5* | 78.04 | **29.09** |
| ApproxDA, gate=learn | $\approx$0.5* | 77.93 | 33.96 |

*Gate collapsed to $g\approx0.5$ throughout training; see Section V-A. **Bold**: best; underline: second best per column.

TABLE IX
CROSS-TASK APPROXIMATION SAFETY VS. GLOBAL CONTEXT REQUIREMENT

| Dataset | GCR | DA-TransUNet† | Best ApproxDA | $\Delta$ DSC |
|---|---|---|---|---|
| Synapse | High | 79.80% | 78.64% (gate=pam) | $-1.16\%$ |
| Kvasir-SEG | Low | 88.47% | 89.24% (gate=learn) | $+0.77\%$ |
| ISIC 2018 | Low | 88.88% | *pend.* | *pend.* |

†Paper-reported results [7]. Kvasir/ISIC baselines use Adam, 500/50ep, 75/25 split; ApproxDA uses SGD, 300ep, 80/20 split. Direction of $\Delta$DSC is meaningful despite this setup difference.

further isolate whether windowing or low-rank projection is the primary source of error on high-GCR tasks.

### D. RQ3: Can Adaptive Routing Compensate for High-GCR Requirements?

To evaluate the effectiveness of adaptive routing, we analyze the routing coefficient throughout inference together with its correlation with branch outputs. Table VIII reports the gate mode ablation results on Synapse.

Surprisingly, the routing coefficient remains nearly constant across different datasets, network stages, and approximation configurations. The learnable gate (gate=learn, 77.93% DSC) ranks last among all configurations—below both PAM-only (78.64%) and CAM-only (78.26%). The collapsed 50/50 blend is weaker than either single-branch fixed mode. Detailed analysis of this collapse phenomenon is presented in Section V-A.
**Answer to RQ3:** No—adaptive routing degenerates: the gate collapses to $g\approx0.5$ regardless of dataset or window size, performing worse than both fixed single-branch modes.

### E. RQ4: Does Approximation Safety Reverse Across GCR Regimes?

Finally, we compare the behavior of ApproxDA across datasets exhibiting different contextual characteristics. Table IX summarizes the cross-task comparison. The direction of approximation's effect reverses across tasks: $-1.16\%$ DSC on the high-GCR Synapse benchmark versus $+0.77\%$ DSC on the low-GCR Kvasir-SEG benchmark.

ApproxDA consistently benefits segmentation tasks dominated by local appearance and boundary information, whereas segmentation performance decreases on tasks requiring long-range anatomical reasoning. Figure 2 illustrates this context-dependent pattern. These observations motivate the analysis presented in Section V.

Fig. 5. Qualitative segmentation comparison. **Top (Synapse):** DA-TransUNet preserves multi-organ boundaries while ApproxDA shows degradation on inter-organ regions, illustrating high-GCR sensitivity. **Bottom (Kvasir-SEG):** ApproxDA matches or exceeds DA-TransUNet on polyp boundaries, illustrating low-GCR tolerance. [Generated after experiments complete.]

**Answer to RQ4:** Yes—task context strongly governs approximation safety. The same configuration yields opposite effects depending on GCR: $-1.16\%$ DSC on high-GCR Synapse vs. $+0.77\%$ DSC on low-GCR Kvasir-SEG, confirming GCR as the governing criterion.

## V. DISCUSSION

### A. Observation 1: Gate Collapse

**Derivation.** Given fused output $\mathbf{F} = g\mathbf{F}_{\mathrm{P}} + (1-g)\mathbf{F}_{\mathrm{C}}$, the gate gradient is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_{\mathrm{P}} - \mathbf{F}_{\mathrm{C}}) . \tag{5}$$

At initialization $g\approx0.5$, both branches receive equal gradient $\frac{1}{2}\partial\mathcal{L}/\partial\mathbf{F}$. Without incentive to differentiate, $\mathbf{F}_{\mathrm{P}} \approx \mathbf{F}_{\mathrm{C}}$, so Eq. (5)$\approx$ $\mathbf{0}$. This is self-reinforcing: equal gradients $\to$ convergent representations $\to$ near-zero gate gradient $\to$ $g$ fixed at 0.5.

**Empirical confirmation.** After 300 epochs, the gate range at $M$=7 is $[0.4993, 0.5010]$ with $\Delta g$=0.0017. At $M$=14: $g\approx0.5002$ with $\Delta g\approx0.0000$. Collapse is identical at both window sizes, confirming it is a fundamental property of symmetric two-branch gating and not a consequence of limited receptive field. Figure 6 visualizes the collapse trajectory.

**Design implication.** Symmetric two-branch gating is an unstable fixed point that any dual-attention architecture will converge to without explicit symmetry breaking. Future designs should initialize branches asymmetrically, use diversity-encouraging loss terms, or replace the gate with a non-symmetric routing mechanism to escape this degenerate equilibrium.

### B. Observation 2: Task-Dependent Approximation Safety

Although ApproxDA adopts the same approximation operators across all experiments, its effectiveness varies considerably among different segmentation datasets. ApproxDA consistently underperforms the original dual-attention module

Fig. 6. Gate value $g$ vs. training epoch for $M{=}7$ and $M{=}14$. Both converge to $g{\approx}0.5$ by epoch 100, independent of window size. Shaded band shows gate range across network blocks. [Generated from training logs after experiments complete.]

Fig. 7. Gate value vs. per-block input entropy scatter (300-epoch checkpoint, Synapse validation). Gate values cluster tightly at $g{\approx}0.5$ regardless of input entropy, confirming no sample-adaptive routing occurs. [Generated from inference analysis.]

on Synapse, while achieving competitive or improved performance on Kvasir.

We attribute this phenomenon to the different global contextual requirements of the underlying segmentation tasks. Multi-organ abdominal CT segmentation requires reasoning over long-range anatomical relationships among multiple organs. Aggressive attention approximation inevitably weakens such global dependency modeling, leading to performance degradation.

In contrast, polyp and skin lesion segmentation are primarily determined by local appearance, boundary contrast, and texture consistency. Consequently, restricting attention to local regions introduces only limited information loss while simultaneously acting as an implicit regularizer, which explains the improved generalization observed on Kvasir.

### C. GCR and Approximation Safety

We formally define two complementary task-level properties:

**Global Context Requirement (GCR):** The degree to which segmentation depends on long-range spatial dependencies between distant anatomical elements. GCR is intrinsic to the task, independent of the model architecture. High-GCR tasks (multi-organ CT) require cross-organ anatomical reasoning—liver position relative to spleen, aorta relative to vena cava—where disrupting global spatial interaction degrades delineation. Low-GCR tasks (polyp, skin lesion) are dominated by local boundary contrast and texture, with minimal dependence on distant context.

**Approximation Safety:** The degree to which spatial attention approximation can be introduced without degrading segmentation accuracy on a given task.

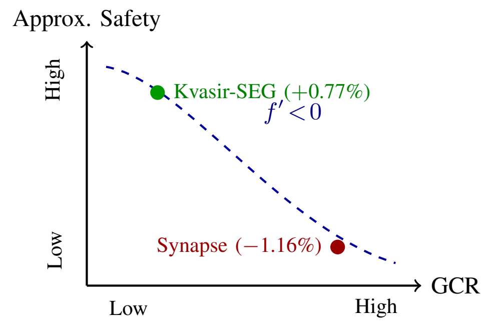


Fig. 8. Conceptual relationship between GCR and approximation safety (Eq. (6)). The dashed curve is hypothetical, not fitted; confirmed data points for Kvasir-SEG (low GCR) and Synapse (high GCR) are consistent with the monotonically decreasing trend. ISIC 2018 result pending.

Our central hypothesis is an inverse relationship between these properties:

$$\text{Approximation Safety} \approx f(\text{GCR}), \quad f'(\cdot) < 0. \tag{6}$$

Equation (6) is a conceptual claim, not an analytic proof. It summarizes the empirical pattern: the same ApproxDA configuration achieves $+0.77\%$ DSC on low-GCR Kvasir-SEG and $-1.16\%$ on high-GCR Synapse—opposite effects consistent with a monotonically decreasing relationship. Figure 8 plots the confirmed data points against this hypothesis.

Attention approximation should not be regarded as a binary design choice but as a space whose safety depends on GCR. Different approximation configurations (Figure 4) suit different GCR regimes. The gate=learn improvement on Kvasir ($+0.77\%$) further suggests that on low-GCR tasks, the implicit regularization of a collapsed 50/50 gate may confer a generalization benefit that PAM-only routing does not.

### D. Design Principles

The GCR framework and the gate collapse finding together provide three practical principles for efficient attention design in medical image segmentation.

First, attention approximation should be selected according to the contextual characteristics of the target segmentation task, rather than applied uniformly across all medical imaging applications.

Second, adaptive routing mechanisms require sufficient feature diversity between attention branches; otherwise, gradient symmetry may cause routing collapse regardless of window size or rank.

Finally, future efficient attention architectures should explicitly consider approximation safety together with computational efficiency, instead of focusing exclusively on reducing theoretical complexity. Figure 9 summarizes these guidelines as a practical decision tree.

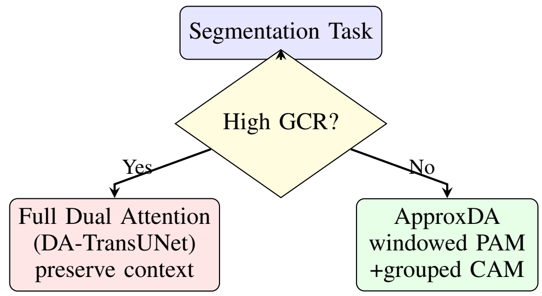


Fig. 9. Practical design guideline derived from GCR analysis. Tasks requiring cross-organ anatomical reasoning (high GCR) should retain full dual attention. Local-structure tasks (low GCR) can safely use ApproxDA, reducing per-GPU VRAM by 44% with no accuracy penalty.

## VI. Conclusion

This paper presented ApproxDA, a controllable approximation formulation derived from the dual-attention module of DA-TransUNet [7]. Rather than proposing another efficient attention operator, ApproxDA was designed to systematically investigate the effectiveness of attention approximation under different medical image segmentation tasks.

Extensive experiments demonstrate that attention approximation is fundamentally context dependent. While approximation remains effective for local-structure-dominant segmentation tasks such as polyp segmentation (ApproxDA achieves 89.24% DSC on Kvasir-SEG, +0.77% over the full-attention baseline), it consistently degrades performance on tasks requiring extensive global anatomical reasoning such as multi-organ CT segmentation ($-1.16\%$ DSC on Synapse).

Furthermore, our analysis reveals that the conventional learnable routing mechanism naturally collapses because of gradient symmetry, causing adaptive routing to degenerate into an almost fixed fusion strategy. This collapse is independent of window size or rank, representing a fundamental limitation of symmetric two-branch gating rather than a windowing artifact.

Together, these findings establish Global Context Requirement (GCR) as a practical criterion for predicting whether attention approximation will be beneficial or harmful for a given segmentation task—shifting the research question from *how to approximate* to *when to approximate*.

Future efficient attention research should move beyond designing increasingly lightweight attention operators toward understanding when approximation is fundamentally appropriate. We hope this study provides practical guidelines for designing future efficient attention architectures in medical image segmentation. In future work, we plan to investigate routing mechanisms that explicitly preserve branch diversity, as well as broader approximation strategies for foundation medical vision models.

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